Germany: energy security, economic opportunity and partnership priorities

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AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- 5 of the 30 placed sectors of Germany sit in the two low-energy-security corners. Its lowest Energy Security percentiles are in downstream nuclear and downstream oil, where its consumption and energy-access measures pull the scores down.[1]
- Its highest Economic Opportunity percentiles are in downstream green hydrogen and midstream gas, where it scores higher than 99% and 98% of the other economies scored.[1]
- On the paper's weighting, its top Strategic Index sector among its own sectors is downstream solar, almost tied with midstream green hydrogen. That sector ranks 1st on policy activity but only 15th without the policy term.[1,2]
- The Partnership Strength Index rates the United States highest for Germany, as a reliable supplier. It is also a priority partner with strong ties today, unlike China, which is a strong tie today that the index ranks lower.[1]

1. Energy security risks and why

Of the 30 placed sectors of Germany, 5 sit in the two low-energy-security corners: 1 in the Upgrade corner and 4 in the Secure corner, which pairs low opportunity with low energy security.[1]

Its lowest Energy Security percentiles are in downstream nuclear and downstream oil, both in the Secure corner: it scores higher than 0% and 2% of the other 66 and 98 economies scored. In downstream nuclear, its consumption and energy access pull the score down. Its consumption in this sector was 0 GJ per person in 2025, a change of -100.0%. Its growth in use per person to 2035 is not reported and is filled in with the world average; in all, 45.8% of the score's weight rests on filled-in data. In downstream oil, the same categories pull the score down. Its consumption per person changed by -5.2%, to 49.9 GJ per person in 2025; on that change it scores higher than only 7% of the other economies.[1,3]

In upstream oil and upstream gas, its energy imports and reserves pull the score down. Oil production minus consumption was -4.13 EJ in 2025, and no oil reserves are recorded, which the index counts as zero. That oil figure uses production from the U.S. Energy Information Administration's international statistics, as the Energy Institute does not report it separately. For gas, production minus consumption was -2.69 EJ in 2025, and proved reserves were 0.0197 tn m3 in 2020.[3,4]

Its high-priority sector, upstream gas, sits in the Upgrade corner, with Energy Security higher than 27% of the other 97 economies scored. For context, the bilateral trade data record USD 840 m of imports of these goods in 2024: 37.1% from Belgium, 24.1% from the Netherlands and 19.3% from the United States.[1,5]

2. Economic opportunities and why

23 of its placed sectors sit in the two high-opportunity corners. Its highest Economic Opportunity percentiles are in downstream green hydrogen, midstream gas and midstream nuclear: it scores higher than 99%, 98% and 98% of the other 157, 155 and 153 economies scored.[1]

In downstream green hydrogen, consumption and technology demand push the score up. On its domestic demand measure, it scores higher than 100% of the other economies; part of the wider demand category is filled in with the world average.[1]

In midstream gas, trade and cost competitiveness push the score up. It exported USD 11.5 bn of these goods in 2024 and held 7.7% of world exports in 2023. For context, its largest markets in 2024 were the United States (10.8%) and Italy (7.4%).[5,6,7]

In midstream nuclear, the same categories push the score up: it held 10.9% of world exports in 2023, and its revealed comparative advantage was 1.82 in 2024. For context, the United States took 29.7% of these exports in 2024. In downstream solar, it scores higher than 98% of the other 147 economies, lifted by investment and production; it produced 91.6 TWh in 2025.[1,3,5,6,7]

3. Sectoral priorities and the existing policy posture

On the paper's weighting, the Strategic Index ranks downstream solar 1st, midstream green hydrogen 2nd, upstream green hydrogen 3rd, upstream solar 4th and upstream wind 5th among the sectors of Germany. These are almost tied: neighbours are 0.002, 0.007, 0.005 and 0.006 apart. On policy activity, downstream solar ranks 1st, but on data-based priority (the Strategic Index without its policy term) only 15th, so its lead rests on policy. The paper's rule takes the Upgrade corner, which the framework associates with leapfrogging (building domestic capability), as high-priority. For Germany that is upstream gas, ranked 19th on the Strategic Index.[1,2]

Its allied pick in the network design is midstream batteries, almost tied with the runner-up, upstream nuclear (0.007 apart). The pick skips upstream sectors without a resource base, which on the data include upstream green hydrogen, upstream solar and upstream wind.[8]

The Climate Action Tracker rates Germany Insufficient (August 2024), and the International Energy Agency's Policies and Measures database holds 448 records for it, 267 naming a technology. Ranked within Germany, policy activity places upstream green hydrogen, upstream solar and upstream wind in the middle group. Splitting data-based priority and policy activity into thirds, no sector combines top priority with the least policy. Top priority and the most policy coincide in midstream green hydrogen, midstream batteries and midstream solar, while downstream green hydrogen combines bottom priority with the most policy. The high-priority upstream gas sits in the middle third on data-based priority and among the sectors with the least policy. The 7 sectors with bottom priority and the least policy are all in coal, oil or gas.[1,2,9]

Country Insights also lists, as context, Global Trade Alert's industrial-policy measures for Germany, EU-wide ones included.

4. Partnership priorities and why

The Partnership Strength Index scores how much a partner offers as a reliable supplier, an export market and a place to build supply chains together. From the viewpoint of Germany, it rates the United States highest, as a reliable supplier. Its best supplier scores are in downstream nuclear, midstream batteries,

upstream gas, midstream wind and upstream oil. Austria, Italy, the Czech Republic and France follow, ranked 2nd, 3rd, 4th and 5th: Austria and the Czech Republic each as a reliable supplier, Italy and France each as an export market.[1]

The United States is also its 1st trade partner on average across its energy-goods baskets, France its 3rd and Italy its 4th; each is a priority partner with strong ties today. Germany and the United States both belong to the G7, NATO and the Minerals Security Partnership. Austria and the Czech Republic rank lower as trade partners, at 9th and 11th.[1,5,10]

China is its 2nd trade partner, but the index ranks it 51st: it is a strong tie today that the index ranks lower. On the index it counts as a development partner, and in votes at the United Nations, Germany and China sit in opposite blocs.[1,5,11]

As a partner to the coalition, Germany ranks 2nd of 192 economies. In the allied network design, its derived shares of coalition production are largest in midstream nuclear (5.4%) and midstream green hydrogen (5.1%).[1,8]

5. Data notes and caveats

See the Country Insights page (12 October 2026 baseline) for each source.

- Scores compare Germany with other economies in the same sector; they are not measures of risk or profit in absolute terms.
- All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings.
- Some figures behind these scores are filled in where the source has no value: with zero where a source lists every holder, otherwise with the world average.
- Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order.
- Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up.
- In the allied network design the European Union is one producer; Germany's shares are derived from its weight within the EU.
- Trade figures count trade with other EU members.
- Germany's counts leave out the EU's own 438 International Energy Agency policy records.

Sources

1. Energy Security, Economic Opportunity, Strategic Index and Partnership Strength Index, paper baseline (Opportunity Security Indices pipeline); CC BY 4.0 (the paper's indices).
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8. Allied network design (the paper's model design of coalition production shares).
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10. Economic-security agreements: G7, NATO, Indo-Pacific Economic Framework, Minerals Security Partnership (and Forum), U.S. free-trade agreements (compiled by the authors from official membership lists).
11. Bailey, M., Strezhnev, A. and Voeten, E., United Nations General Assembly Voting Data, ideal points (Harvard Dataverse V39.0) (sessions to 2023); attribution; https://dataverse.harvard.edu/dataset.xhtml?persistentId=hdl:1902.1/12379.

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